

Assessing imputation techniques for missing data using statistical models: a simulation approach

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1. Introduction & Context

- Easy access to information in the 21st century
- Incomplete datasets can result in biased analyses
- Data loss and missing values are common issues in surveys



1.1 Introduction & Context

Missing Data in:

- Politics
- Health
- Economics

Are there any solutions?

- Listwise or pairwise deletion
- **Data imputation**

1.2 Introduction & Context

Simple imputation

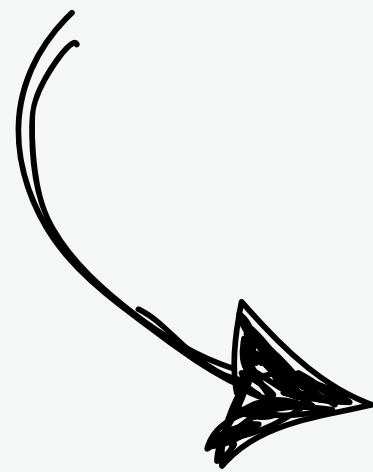
- Mean, median or mode
- Values from other variables with similar characteristics

Advanced imputation

- Statistical modeling (regression)
- Expectation-Maximization
- Machine learning
- Deep Learning-Based

...

About this research



- Investigate missing data imputation through predictive modeling;
- Discuss possibilities on the topic;
- Simulation approach with different scenarios;
- Practical guide.

2. Statistical Regression

- Study of the relationship between two or more variables:

Response variable (Y_i)  Explanatory variable (X_{ki})

- The nature of Y_i suggests the appropriate regression method:
 1. Linear models
 2. Generalized Additive Models (GAM)
 3. Generalized Linear Models (GLM)
 4. Cox regression
 -

2. Statistical Regression

Linear

- Quantitative (discrete or continuous)

Logistic

- Dichotomous: binary
- Multicategorical: nominal or ordinal

Poisson

- Positive discrete (count)

3. Imputation Techniques

Statistical analysis >>> Complete datasets

- What is missing data?
- What are the types of imputation?
- What are the known techniques?



Missing Data

- What is the reason for the data loss?

According to Rubin and Wiley (1987) and Harrell (2015):

- **MCAR** (Missing Completely at Random);
- **MAR** (Missing at Random);
- **MNAR** (Missing Not at Random).

3. Imputation Techniques

SINGLE imputation

- Occur only once:
- Measures of central tendency
 - Values from other similar observations
 - Last or next observed value

MULTIPLE imputation

- An Iterative Process:
- Predictive modeling
 - Predictive Mean Matching (PMM)
 - Markov Chain Monte Carlo (MCMC)

Imputation by Predictive Modeling

- Robustness and consistency
- The nature of the response variable defines which regression will be used
- What are the limitations and disadvantages?

Regression



Linear - quantitative continuous

Dichotomous logistic - qualitative binary (0 or 1)

Poisson - quantitative discrete

How to evaluate the performance of the techniques?

- Decision on the best comparison metric
- Simulation study

Some possible comparison metrics:

- General cases: Means, covariance, and correlations
- In regressions: MSE, R^2 , and AIC

4. Applications & some results

- Exploring different missing data scenarios
- Single vs Multiple Imputation
- Simple to complex methods
- Simulations developed in R software (4.3.2)

**Three main
parts**

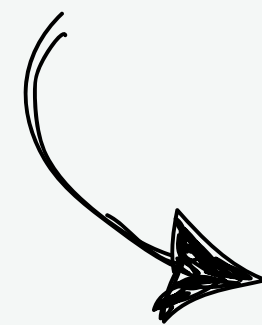


- **Part 1:** Single Imputation in simple scenarios;
- **Part 2:** Single imputation in complex scenarios;
- **Part 3:** Multiple imputation.

Principal database

- Simulated with pseudo-random numbers;
- $n = 200$ observations;

Composed of **6 variables**:



- y, x_2, x_4 : continuous;
- x_3, x_5 : discrete and positive;
- x_1 : dichotomous.

Final predictive model

- Linear regression: $Y \sim X_1 + X_2 + X_3 + X_4 + X_5$
- All assumptions met and consistent.

Measures obtained for the model
fitted to the complete data:

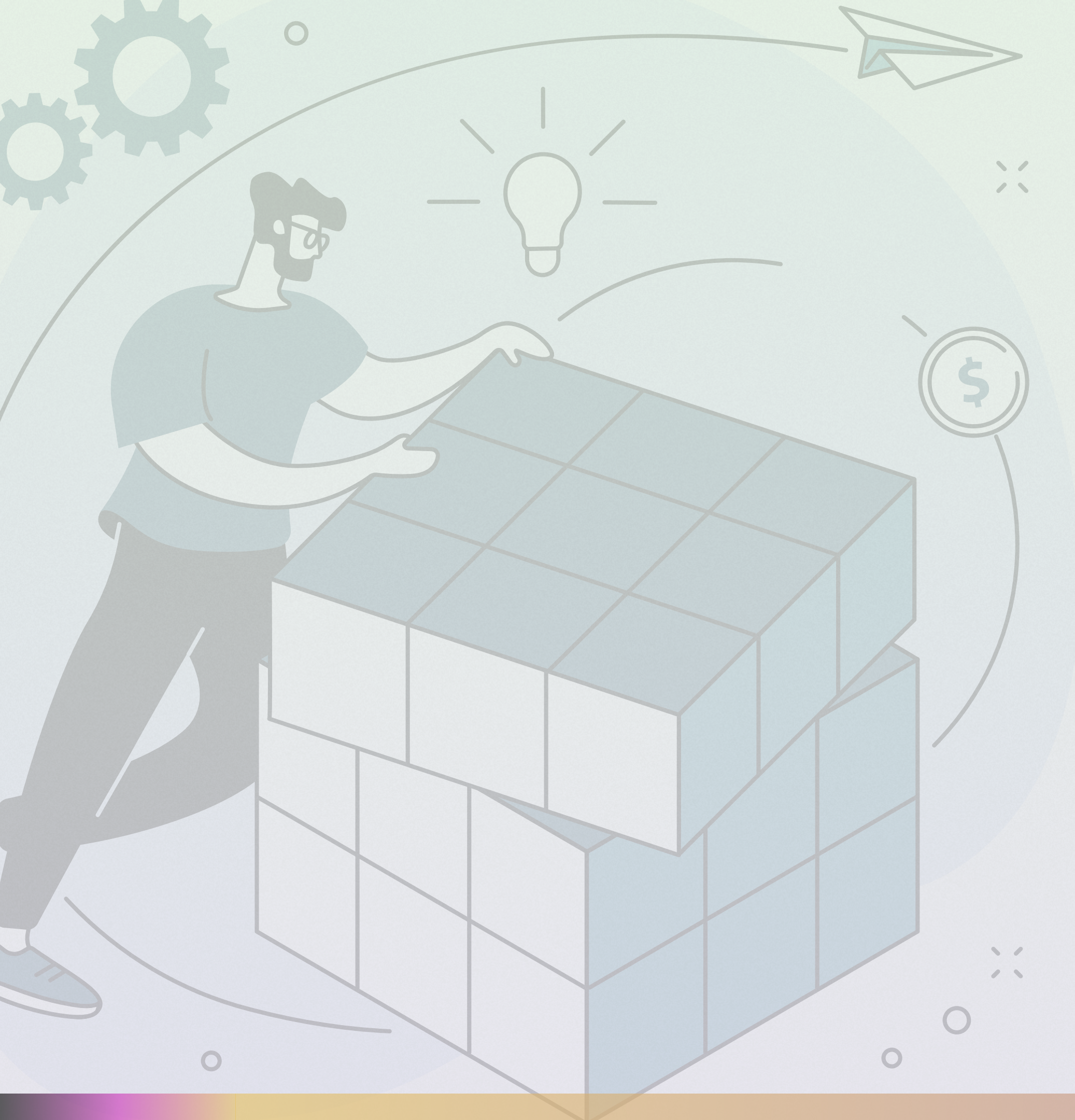
$$\left\{ \begin{array}{l} \text{MSE} = 11.256 \\ R^2 = 0.936 \end{array} \right.$$

Simulation of missing data

- 1 Generate bases with different proportions of missing data in certain variables
- ↓
- 2 5 proportions of missing data (1%, 5%, 10%, 20% and 40%)
- ↓
- 3 300 simulated databases for each proportion
- ↓
- 4 A total of 1500 databases for each technique
- ↓
- 5 Different study scenarios and applications

Comparison criteria

- Final predictive models (MSE and R-squared)
- Descriptive statistics (mean, variance and quartiles)
- Boxplots and density curves;
- Hypothesis tests for comparing samples



4. Applications & some results

Some findings:

Part 3 of the study

4. Applications & some results: about **PART 3**

- **Single vs Multiple** imputation

- **Two variables** with missing data:

- 1 **x1** (dichotomous): 20% missing data
- 2 **x5** (count): all proportions of missing data

- Imputation by **predictive modeling**, regression only: linear for x5; and logistic for x1.

- Iterative process using the MICE package in R:

- 1 x1: 12 imputations for each missing data point
- 2 x5: 24 imputations for each missing data point

- Comparison between single and multiple imputation (a total of 4 scenarios)

4. Applications & some results: about **PART 3**

1 SINGLE IMPUTATION

- x_1 is disregarded from the fits:

Poisson regression: $x_5 \quad x_2$

Linear regression: $x_5 \quad x_2$

2 SINGLE IMPUTATION

- First impute x_1 , then x_5 :

Logistic regression: $x_1 \quad x_2 + x_4$

$\left\{ \begin{array}{l} \text{Poisson regression: } x_5 \quad x_1 + x_2 \\ \text{Linear regression: } x_5 \quad x_1 + x_2 \end{array} \right.$

3 MULTIPLE IMPUTATION

- First impute x_1 , then x_5 :

1st: Logistic regression: $x_1 \quad x_2 + x_4$

2nd: Linear regression: $x_5 \quad x_1 + x_2$

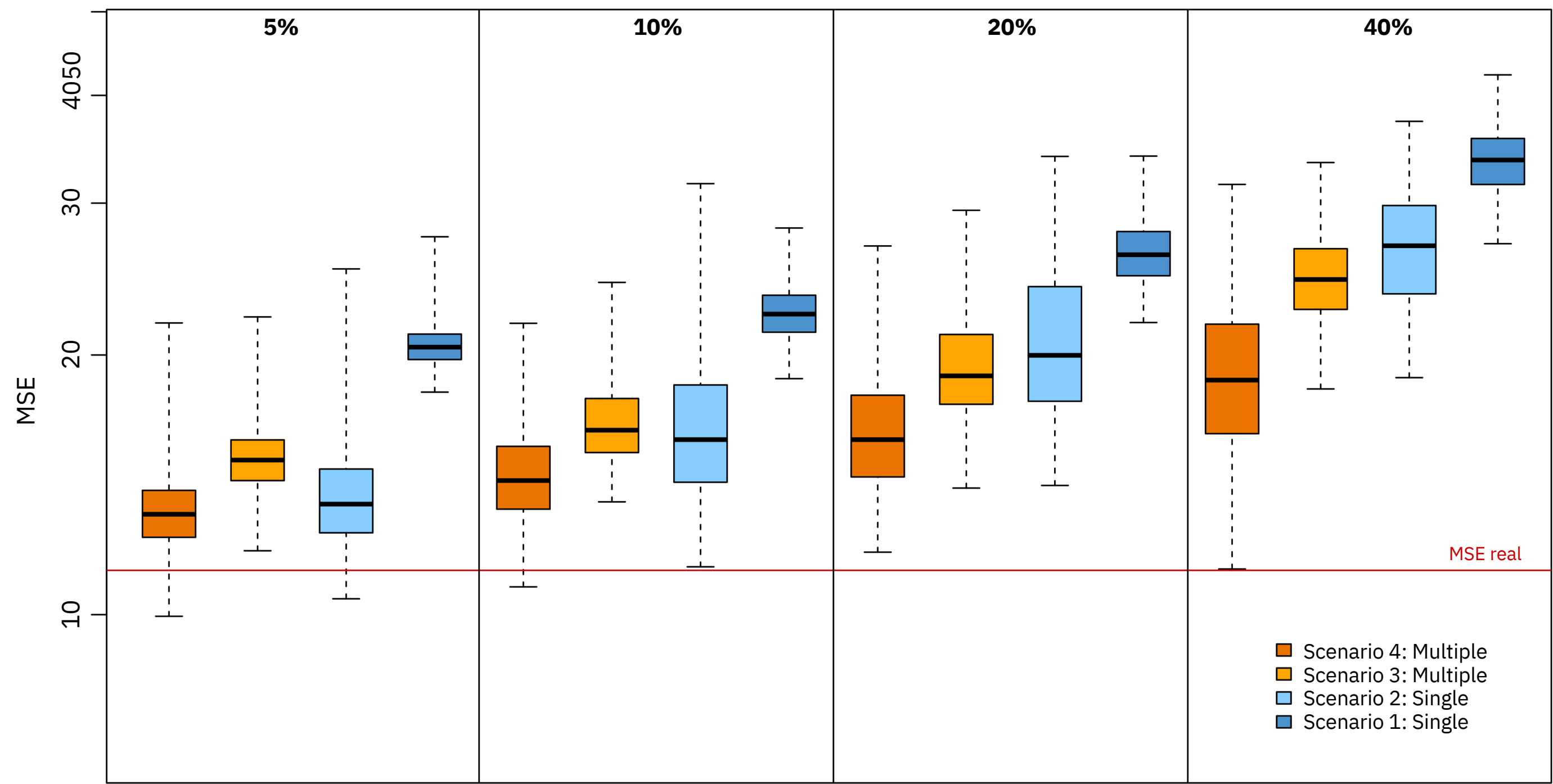
4 MULTIPLE IMPUTATION

- Data from x_1 and x_5 are imputed simultaneously:

Logistic regression: $x_1 \quad x_2 + x_4 + x_5$

Linear regression: $x_5 \quad x_1 + x_2$

Figure 1: First result of part 3. Boxplots referring to the MSEs obtained in scenarios 1 to 4, to represent the results of techniques using only **LINEAR MODELS** to impute x5 values.



Scenario 4: (multiple imputation) x1 and x5 data are imputed simultaneously.

Scenario 3: (multiple imputation) first impute x1, then x5.

Scenario 2: (single imputation) variable x1 is indispensable for model adjustments.

Scenario 1: (single imputation) ignores variable x1 from the model adjustments.

Results after x5 data imputed by **LINEAR MODELS**

- **Best:** multiple imputation, in general;
- **Low proportion:** single imputation (scenario 2) similar to simultaneous multiple imputation (scenario 4);
- The performance of multiple imputation techniques **improves** as the amount of missing data increases, compared to other methods
- **High proportion:** non-simultaneous multiple imputation (scenario 3) is better than single (scenario 2).

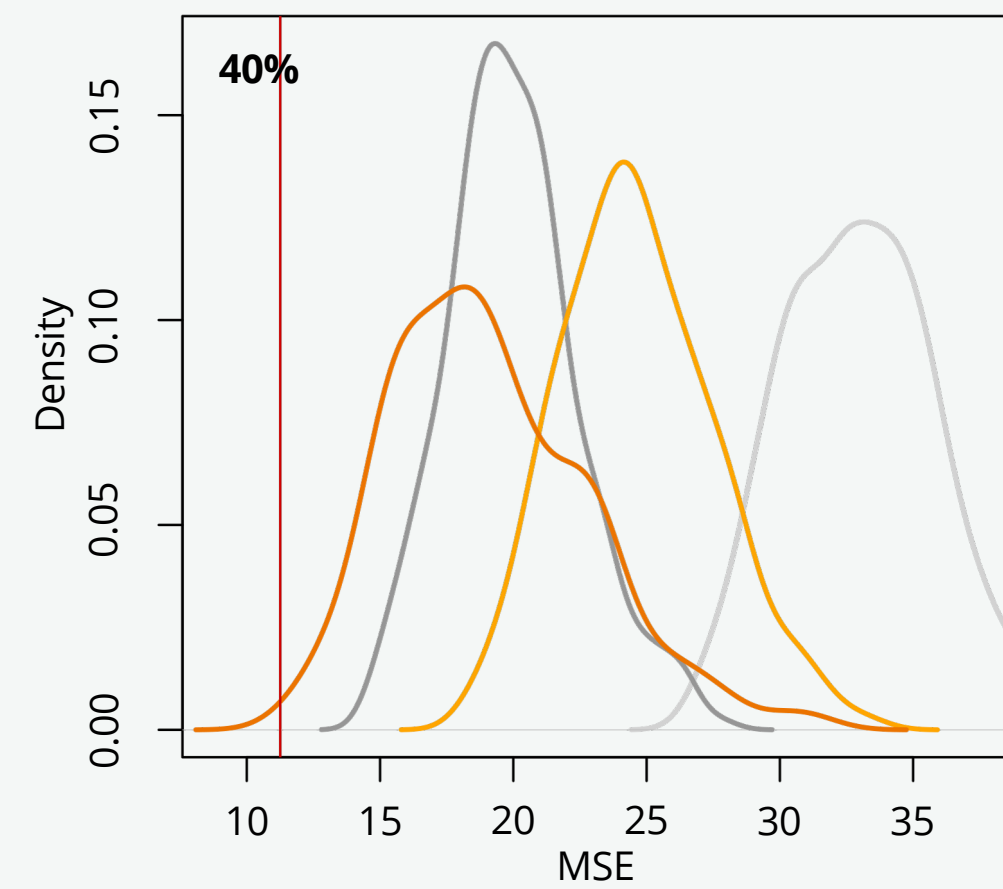
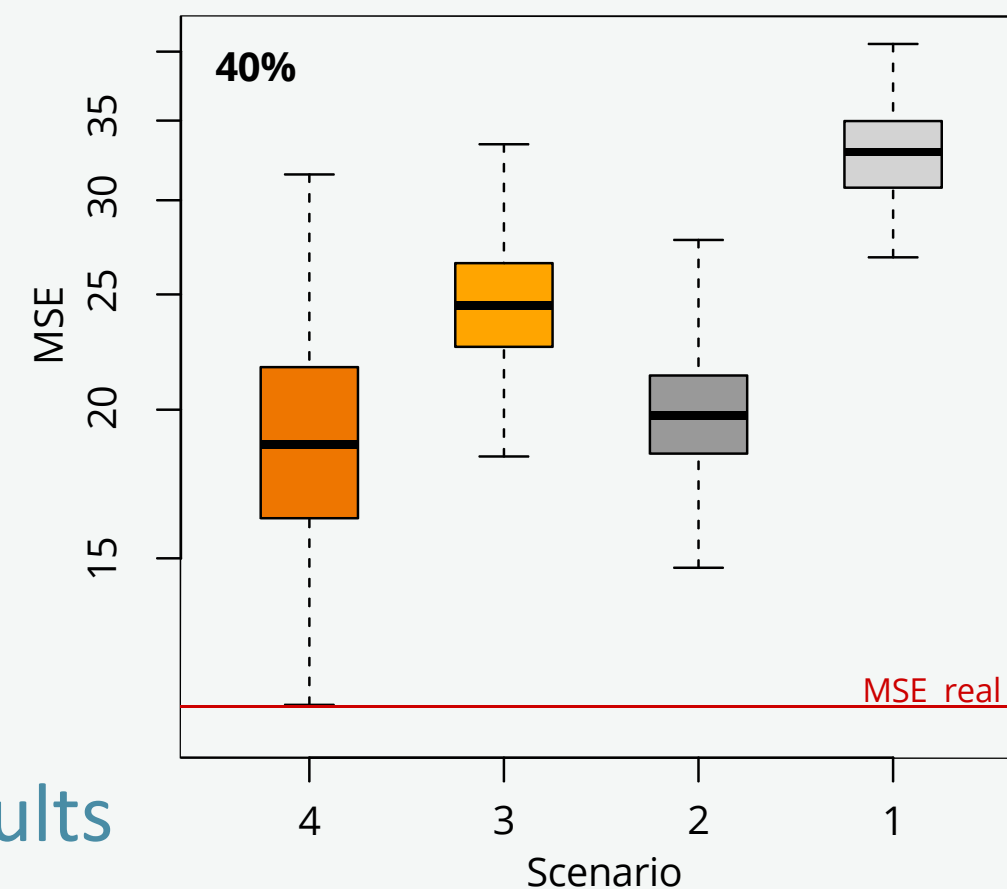
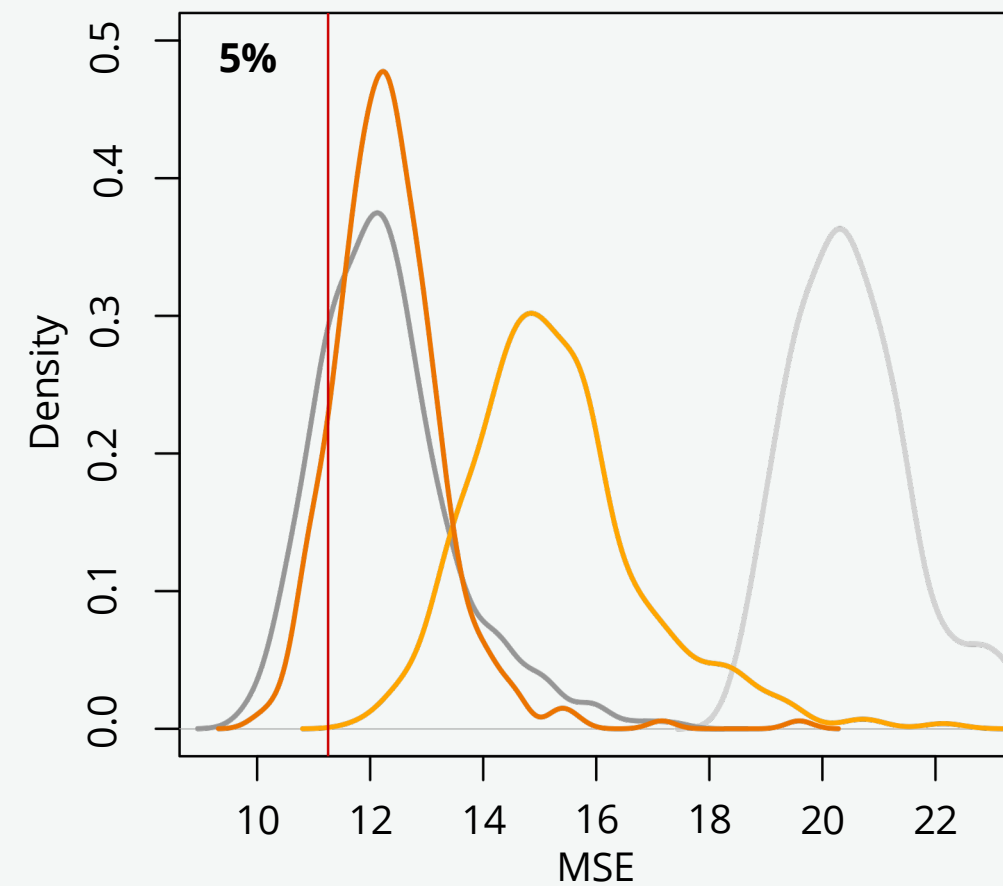
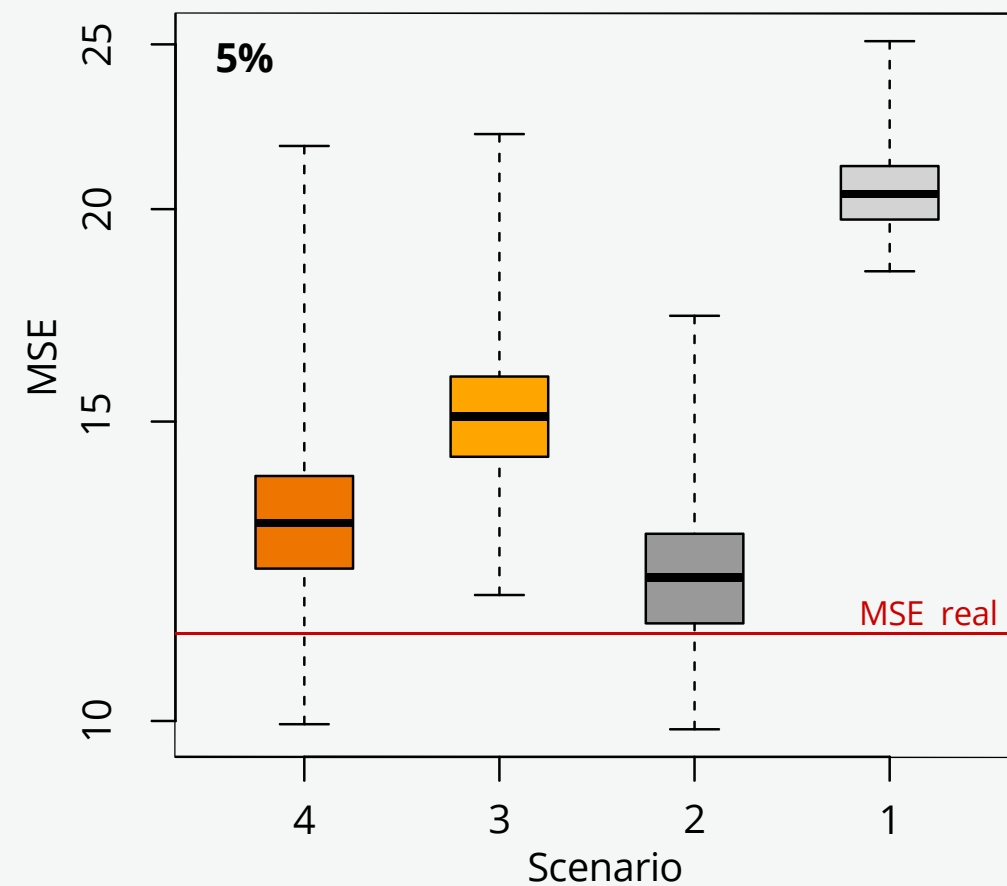
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Scenario 1: (single imputation) ignores variable x1 from the model adjustments.

Figure 2: Second result of part 3. Single imputation with **POISSON** models (scenarios 1 and 2); and multiple imputation with **LINEAR** models (scenarios 3 and 4).



- **Low proportion:** single imputation (Poisson) of scenario 2 proved to be the most efficient;
- **High proportion:** multiple (linear) imputation of scenario 4 proved to be the most efficient

• x5: multiple - **LINEAR**

≈

• x5: single - **POISSON**

Table 1: Descriptive measures for MSEs obtained in Parts 2 and 3 of the study.

Proporção	Cenário	Imputação	Técnica	Média	Variância	Q _{2.5%}	Mediana	Q _{97.5%}
1%	4	Múltipla	Reg Linear	12.345	1.037	10.776	12.265	14.484
	3	Múltipla	Reg Linear	14.129	0.972	12.403	14.074	16.140
	2	Única	Reg Linear	11.708	1.144	10.246	11.583	13.798
	1	Única	Reg Linear	19.111	0.286	18.363	18.994	20.356
	2	Única	Reg Poisson	11.480	0.510	10.050	11.471	12.926
	1	Única	Reg Poisson	19.098	0.229	18.411	19.015	20.207
5%	4	Múltipla	Reg Linear	13.323	2.674	10.954	13.072	17.480
	3	Múltipla	Reg Linear	15.301	2.391	12.899	15.108	19.031
	2	Única	Reg Linear	14.103	7.175	11.085	13.432	22.601
	1	Única	Reg Linear	20.647	1.898	18.839	20.426	24.263
	2	Única	Reg Poisson	12.283	1.489	10.450	12.142	15.152
	1	Única	Reg Poisson	20.598	1.491	18.848	20.413	23.599
10%	4	Múltipla	Reg Linear	14.646	4.128	11.614	14.303	19.181
	3	Múltipla	Reg Linear	16.758	3.617	13.895	16.363	21.109
	2	Única	Reg Linear	16.837	12.654	12.372	15.955	24.786
	1	Única	Reg Linear	22.538	3.294	19.898	22.308	26.997
	2	Única	Reg Poisson	13.600	2.491	11.187	13.386	17.126
	1	Única	Reg Poisson	22.353	2.735	19.737	22.169	26.305
20%	4	Múltipla	Reg Linear	16.468	7.155	12.775	15.956	22.911
	3	Múltipla	Reg Linear	19.421	6.505	14.985	18.917	24.780
	2	Única	Reg Linear	20.898	16.909	15.105	19.985	30.016
	1	Única	Reg Linear	26.469	5.760	22.479	26.138	32.150
	2	Única	Reg Poisson	15.806	4.006	12.399	15.679	19.612
	1	Única	Reg Poisson	26.075	4.593	22.397	25.900	30.428
40%	4	Múltipla	Reg Linear	19.112	13.803	13.151	18.705	27.452
	3	Múltipla	Reg Linear	24.645	8.221	19.738	24.469	30.775
	2	Única	Reg Linear	26.732	15.600	19.581	26.771	34.349
	1	Única	Reg Linear	33.699	9.190	28.293	33.662	39.721
	2	Única	Reg Poisson	20.013	6.029	15.566	19.781	25.590
	1	Única	Reg Poisson	32.980	7.968	27.951	32.951	38.811

5. Practical Guide

- Type of imputation: single or multiple?
- Selection of variables and processes: diagnostic analysis.
- Some **packages** and functions in R:

MICE

Hmisc

mlr

missForest

Amelia

5. Conclusion

1. Spotlight on simultaneous multiple imputation;
2. Single imputation with the correct technique is sufficient for cases of low missing data;
3. Disregarding missing data proved to be the worst solution;
4. Three main factors influence the performance of the techniques:
 - the amount of missing data;
 - the number of incomplete variables;
 - the nature of these variables.
5. Future work: explore new scenarios and techniques, with increasingly specialized guides.

References

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